

0900-1200 hrs

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DJ Exam Hall

EXAMS OFFICE
USE ONLY

University of the Witwatersrand, Johannesburg

Course or topic No(s)

COMS4033A/7044A

Course or topic name(s)
Paper Number & title

Artificial Intelligence

Examination to be held
during the month(s) of

November, 2023

Year of study

Degrees/Diplomas for
which this course is
prescribed

Faculties presenting can-
didates

Science

Internal examiner(s)

Dr Steven James
x-76157

External examiner(s)

Prof. George Konidaris (Brown)

Special materials

None

Time allowance

3 Hours

Instructions to candidates

70 Marks available. 70 marks = 100%.
Answer all questions. This is a closed book exam.
This exam consists of 14 pages.

Artificial Intelligence

COMS4033A/7044A

Student Number: **MEMO** _____ Row: _____ Seat: _____

For marking purposes only

Question 1	
Question 2	
Question 3	
Question 4	
Question 5	
Total	

Instructions

- Answer all questions in pen. **Do not write in pencil.**
- This test consists of 14 pages. Ensure that you are not missing any pages.
- This is a **closed-book** test: you may not consult any written material or notes.
- You are allocated 3 hours to complete this exam.
- There are 5 questions and 70 marks available.
- Ensure your cellphone is switched off.

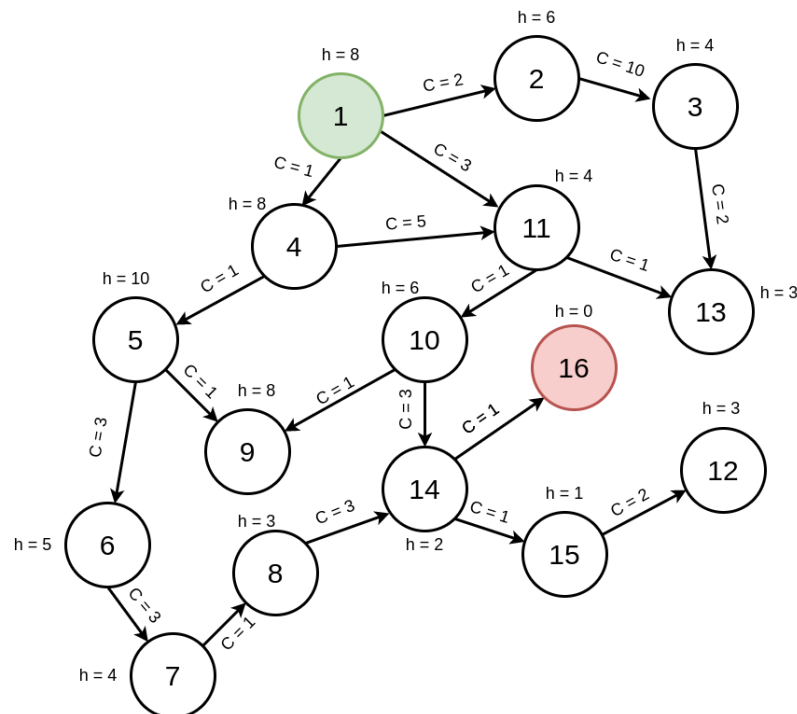
Question 1**Agents and Search****[18 Marks]**

1. What is the difference between the strong AI hypothesis and the weak AI hypothesis? [1]

Weak: Agent can be made to act intelligent (as a human)

Strong: Agent can be made to act as intelligently as a human and is thinking/has mind

2. Let C be the cost of traversing an edge between nodes, and let h be the heuristic estimate of a given node to the goal. Consider the graph below, where nodes are labelled with integers and directed edges are indicated by lines with arrows. Each edge is labelled with a cost C , and each node with a heuristic value h .



Assume that we wish to search for the goal node 16 starting at node 1 using the greedy best first search (GBFS) algorithm. Further, assume that if there are ties in the priority queue, they are resolved by removing the node with the smallest label first (e.g. if two nodes 2 and 1 both have priority 10, then 1 will be removed before 2).

Given this, detail each step of the GBFS algorithm. At each iteration, describe what node is popped from the queue and then what nodes are added to the queue subsequently (and their corresponding priority). [4]

Start with 1. Add 2 (p=6), 4 (p=8), 11 (p=4)

1mk

Pop 11. Add 13 (p=3), 10 (p=6)

1mk

Pop 13. Pop 2. Add 3 (p=4)

1mk

Pop 3. Pop 10.

0.5 mk

Add 14 (p=2). Pop 14. Add 16 (p=0).

0.5mk

Pop 16. Finish

(Give marks if mistake made but rest is correct wrt that mistake)

3. Consider the game of reconnaissance blind chess as described in the assignment. We can categorise this task as follows:

- | | | |
|--|------------|----------|
| (a) partially observable or fully observable | partial | |
| (b) deterministic or stochastic | stochastic | |
| (c) episodic or sequential | episodic | 0.5 each |
| (d) static or dynamic | sequential | |
| (e) discrete or continuous | discrete | |
| (f) single agent or multiagent | multi | |

For each option above, circle the correct one. [3]

4. List one advantage of depth-first search over breadth-first search? [1]

Better memory complexity (linear vs exp)

5. Consider the following payoff matrix representing a game between two players A (the “row” player) and B (the “column” player).

	B_1	B_2	B_3	B_4
A_1	(1, 7)	(2, 5)	(7, 2)	(0, 1)
A_2	(5, 2)	(3, 3)	(5, 2)	(3, 1)
A_3	(7, 0)	(2, 5)	(0, 4)	(1, 1)
A_4	(0, 0)	(0, 2)	(0, 0)	(0, -1)

- (a) Execute the *maximin* algorithm on the payoff matrix to find the outcome, beginning with player *A*. At each step, list the strategy selected by the relevant player, as well as the final outcome. [2]

A would select A2 (min guaranteed = 3)

B would select B2 as a response. Outcome is (3,3)

- (b) Repeat the above, but now begin with player *B*. What is the outcome of this procedure? [1]

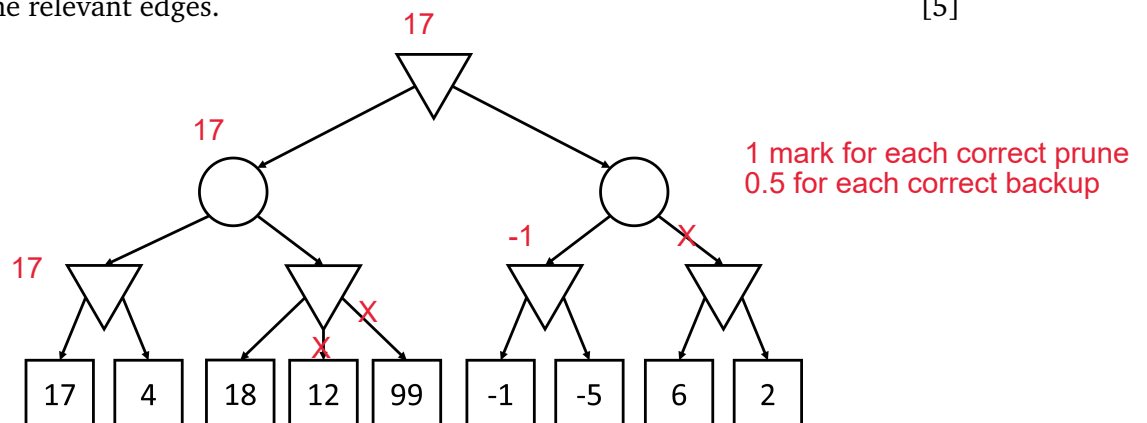
B would select B2 (min guaranteed = 2)

A would respond with A2. Outcome is (3,3)

- (c) What can be said about the strategies produced by the *maximin* algorithm for both players? Motivate your answer. [1]

A2,B2 is a nash equilibrium. No player can do better by unilaterally switching

6. Consider the game tree below. Triangles represent the maximising player, circles the minimising player and rectangles terminal nodes. Execute alpha-beta pruning on the tree. Annotate the tree by writing down the value backed up to each node in the tree (including the root), and indicate where pruning takes place by crossing out the relevant edges. [5]



Question 2 Knowledge Representation [9 Marks]

1. Consider a knowledge base with three objects A, B and C. Instantiate the following sentence: $\forall x, \text{Human}(x) \Rightarrow \text{Mortal}(x)$. [1]

Human(A) -> Mortal(A)

Human(B) -> Mortal (B)

Human(C) -> Mortal(C)

2. Alice wishes to encode the following sentence in first-order logic: "John does not love anyone" She writes this as $\exists x \neg \text{love}(\text{John}, x)$. Her friend Bob disagrees with her and thinks it should be written as $\neg \forall x \text{love}(\text{John}, x)$. Eve disagrees with them both and thinks it should be $\neg \exists x \text{love}(\text{John}, x)$. Explain who is correct and provide a reason why the other two are incorrect. [3]

Eve is correct: her statement is there does not exist even one person who is loved by John.

Alice is wrong: hers says there exists at least one person who John does not love

but could be others that he does.

Bob is wrong: same as Alice. John does not love everyone. So John could love some, but not others

3. Assume the probability of having some disease is 0.0005, and a test for this disease is 99% accurate. What is the probability one has this disease if one tests positive for it? [2]

$$P(D|T) = P(T|D) * P(D) / P(T)$$

$$= 0.99 * 0.0005 / P(T) = 0.0472$$

$$P(T) = P(T|D)P(D) + P(T|\text{not } D)P(\text{not } D) = 0.99*0.0005 + 0.01*0.9995$$

4. A patient has been suffering from shortness of breath (called dyspnoea) and visits the doctor, worried that he has lung cancer. The doctor knows that the chance of the patient having cancer (or some other disease) is affected by whether he is a smoker or has been exposed to air pollution. A positive X-ray would indicate lung cancer and would explain the patient's dyspnoea. We wish to model this using a Bayesian network with the following random variables: whether the patient has been exposed to pollution (*pollution*), whether they are a smoker (*smoker*), whether they have cancer (*cancer*) or dyspnoea (*dyspnoea*) and whether the X-ray returns a positive cancer diagnosis (*xray*).

- (a) Draw the corresponding Bayesian network for the above three variables. [2]

pollution, smoker are independent. Both point to cancer

Cancer points to xray.

Cancer points to dyspnoea

- (b) Write down the factorisation of joint probability distribution given the Bayesian network's structure in part a). [1]

$$P(\text{joint}) = P(\text{Pollution})P(\text{smoker})P(\text{cancer}|\text{pollution,smoker})P(\text{xray}|\text{cancer})P(\text{dysp}|\text{cancer})$$

(give mark if correct based on what is drawn in a)

Question 3**Planning****[23 Marks]**

1. What is the fundamental difference between search and STRIPS planning? [1]

Search treats states as atomic black boxes. STRIPS adds in structure

2. What is the difference between a PDDL precondition and effect? [2]

Precondition: what must be true for operator to be executable

Effect: how does executing operator change the state of the world

3. List the components that make up a PDDL problem description. [3]

init description

objects

goal test

4. Describe how the FF planner creates a relaxed version of a planning problem [2]

It removes negatives effects and solves the resulting problem to generate a heuristic for the full problem

5. Consider the Blocksworld environment with ten blocks labelled A–J. The current state is given by

```
(clear C) (clear F) (ontable I) (ontable F) (on C E) (on E J)
(on J B) (on B G) (on G H) (on H A) (on A D) (on D I) (handempty))
```

and there is an operator `unstack`, which is defined as follows:

```
(:action unstack
  :parameters (?x ?y)
  :precondition (and (on ?x ?y) (clear ?x) (handempty))
  :effect
  (and (holding ?x)
    (clear ?y)
    (not (clear ?x))
    (not (handempty))
    (not (on ?x ?y))))
```

Write down the new state that results from the execution of `unstack(c, e)`. [3]

(clear F) (ontable I) (ontable F) (on E J)
(on J B) (on B G) (on G H) (on H A) (on A D) (on D I) (holding c) (clear e)

[1 mk for correct deletes, 1 mark for correct adds, 1 mark for copying rest]

6. Under what conditions do we expect that regression planning would outperform forward planning? [1]

When there is a narrow solution path/low branching factor at the goal, but high branching factor at the start

7. Define what is meant by a *macro-action*. [1]

New operator composed of existing operators (chained together)

8. Name two things that make a scheduling problem harder than a planning problem. [2]

Simultaneous actions

Need to account for time

9. What are the five components that make up a Markov decision process? Explain what each of them are. [3]

S - set of states

A - set of actions available

T - transition dynamics

R - reward function

gamma - discount factor

10. In words, define what is meant by the *return*. [1]

Sum of discounted rewards obtained by an agent executing a policy

11. What is the difference between a policy and a plan? [1]

Plan: sequence of actions (open loop)

Policy: specifies what action to take in each state (closed loop)

12. When estimating a value function for a policy, why is Monte Carlo considered to be high variance? [1]

MC uses sum of rewards from rollouts, which depend on trajectory. Each reward is a product of stochasticity

13. What is the difference between policy iteration and prioritised sweeping? [2]

PI loops over all states. PS prioritises updates based on Bellman error/ predicted change in value function. So more relevant states are updated first and value can be propagated faster

Question 4 Reinforcement Learning [8 Marks]

1. Explain how Q-learning overcomes the challenge of having to act greedily with respect to a value function. [1]

Learn state-action value function that specifies value for each state and action

Then maximise to get action at each state

2. Describe what is meant by the exploration-exploitation dilemma. [2]

Agent must take potentially suboptimal actions to try see if there's something better

But must also use what it knows to accumulate reward and act greedily

3. Write down the SARSA update rule. How does this differ from the Q-learning update rule? [3]

$Q(s,a) = Q(s,a) + \alpha * (r + \gamma * Q(s',a') - Q(s,a))$

where $a' = \pi(s')$

In Q learning, $a' = \operatorname{argmax}(Q(s, b))$

4. What is the main difference between early (pre 2000) attempts at function approximation, and function approximation using deep learning (with neural networks?) [1]

Before: designer specifies the features

Now: NN learns the features

5. Describe how the DQN algorithm overcomes the problem training using data that is highly correlated. [1]

Uses a replay buffer to store data and randomly sample from it to break the dependency on using data from successive transitions

Question 5**Machine Learning****[12 Marks]**

1. With regards to regression, describe how overfitting occurs and why it is problematic. [2]

In regression, we might use an overly complex function to exactly fit the data and all its noise. We get 0 training error, but will fail to generalise

2. Describe the basic idea behind k -fold cross validation. [1]

Split data into k chunks. Train on $k-1$ of them and test on the held out one

3. How does a decision tree learning algorithm decide which attribute to test next? [2]

Use information gain. Which attribute will split the data into two subsets whose information is "balanced"

4. Describe how the *bandwidth* parameter in kernel density estimation relates to overfitting and underfitting. [2]

If bandwidth too small, KDE will be sensitive to local changes and overfit
If too large, KDE will take distant points into account too much and underfit

5. Describe how autoencoders can be pretrained for downstream classification tasks. [2]

Pretrain autoencoder to reconstruct input. Then remove layers after the bottleneck and replace them with new ones. Train the entire thing from there.

6. Consider the following datapoints: $x_1 = (2, 10)$, $x_2 = (2, 5)$, $y_1 = (5, 8)$, $y_2 = (7, 5)$, $z_1 = (1, 2)$, $z_2 = (4, 9)$. We wish to perform k -means clustering with a Euclidean distance metric on this data with $k = 3$. The initial centroids are chosen to be x_1, y_1, z_1 . Given this, write down all steps of the algorithm until convergence. Your answer should include distance calculations, as well as any updates to the centroids and the labelling of datapoints to their clusters. [3]

```
% 2.00 10.00 5.00 8.00 1.00 2.00 Cluster New Cluster
% A1 2.00 10.00 0.00 3.61 8.06 1
% A2 2.00 5.00 5.00 4.24 3.16 3
% B1 5.00 8.00 3.61 0.00 7.21 2
% B2 7.00 5.00 7.07 3.61 6.71 2
% C1 1.00 2.00 8.06 7.21 0.00 3
% C2 4.00 9.00 2.24 1.41 7.62 2
```

```
% 2.00 10.00 5.33 7.33 1.50 3.50 Cluster New Cluster
% A1 2.00 10.00 0.00 4.27 6.52 1 1
% A2 2.00 5.00 5.00 4.06 1.58 3 3
% B1 5.00 8.00 3.61 0.75 5.70 2 2
% B2 7.00 5.00 7.07 2.87 5.70 2 2
% C1 1.00 2.00 8.06 6.87 1.58 3 3
% C2 4.00 9.00 2.24 2.13 6.04 2 2
```

Stop

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